

Navigator



Find your way on a large sample from its design file.

The camera sees a few hundred micrometres of a sample that is millimetres across. The Navigator shows the sample's layout (GDS) or a photo of it under the stage, registered to the stage by a few reference points. Click a structure on the layout, read its stage coordinates and move there. It works with any AaltoFlow stage module and has so far been tested against the simulated stage.

Designs: **GDSII** / **OASIS** (read with gdstk, no KLayout needed) or **PNG** / **JPG** / **TIFF** with its real width · stages: **KIM101** (μm), **Thorlabs BSC203** (mm), or any module with X and Y position controls

Module source on GitHub

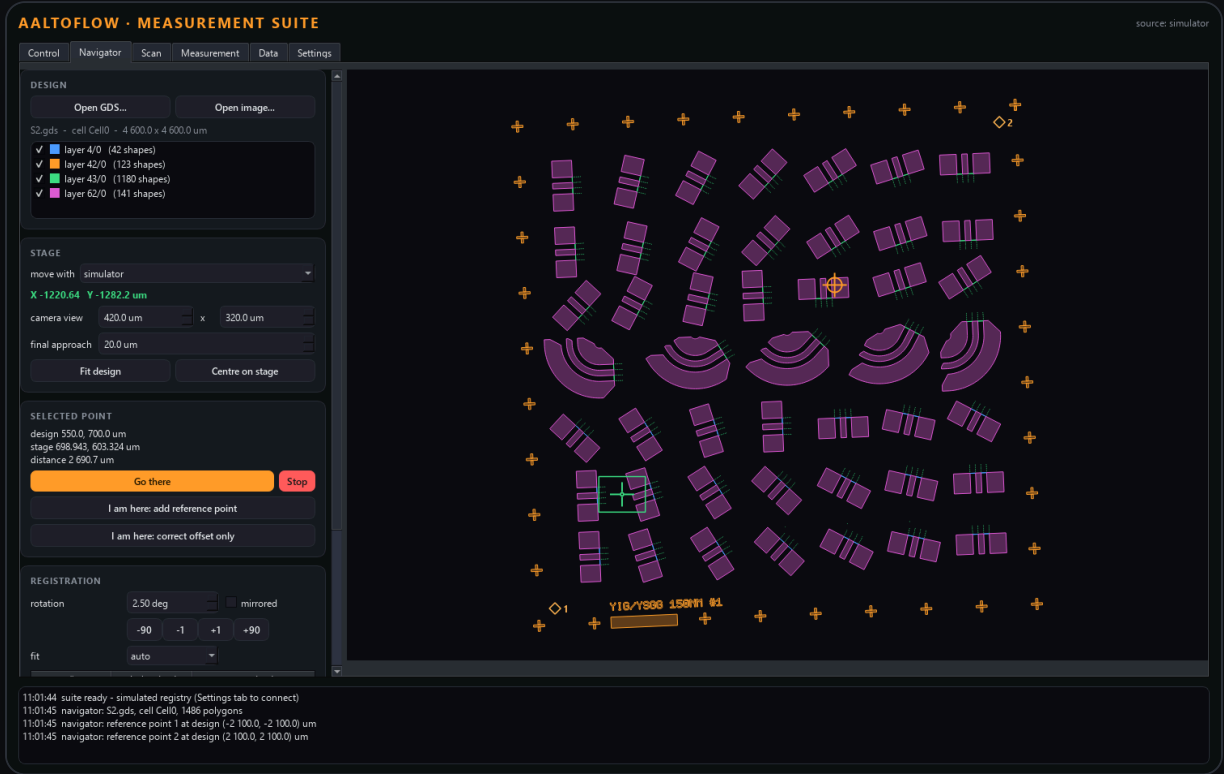
AaltoFlow on GitHub

GDS or image point registration click and go drift correction simulator included

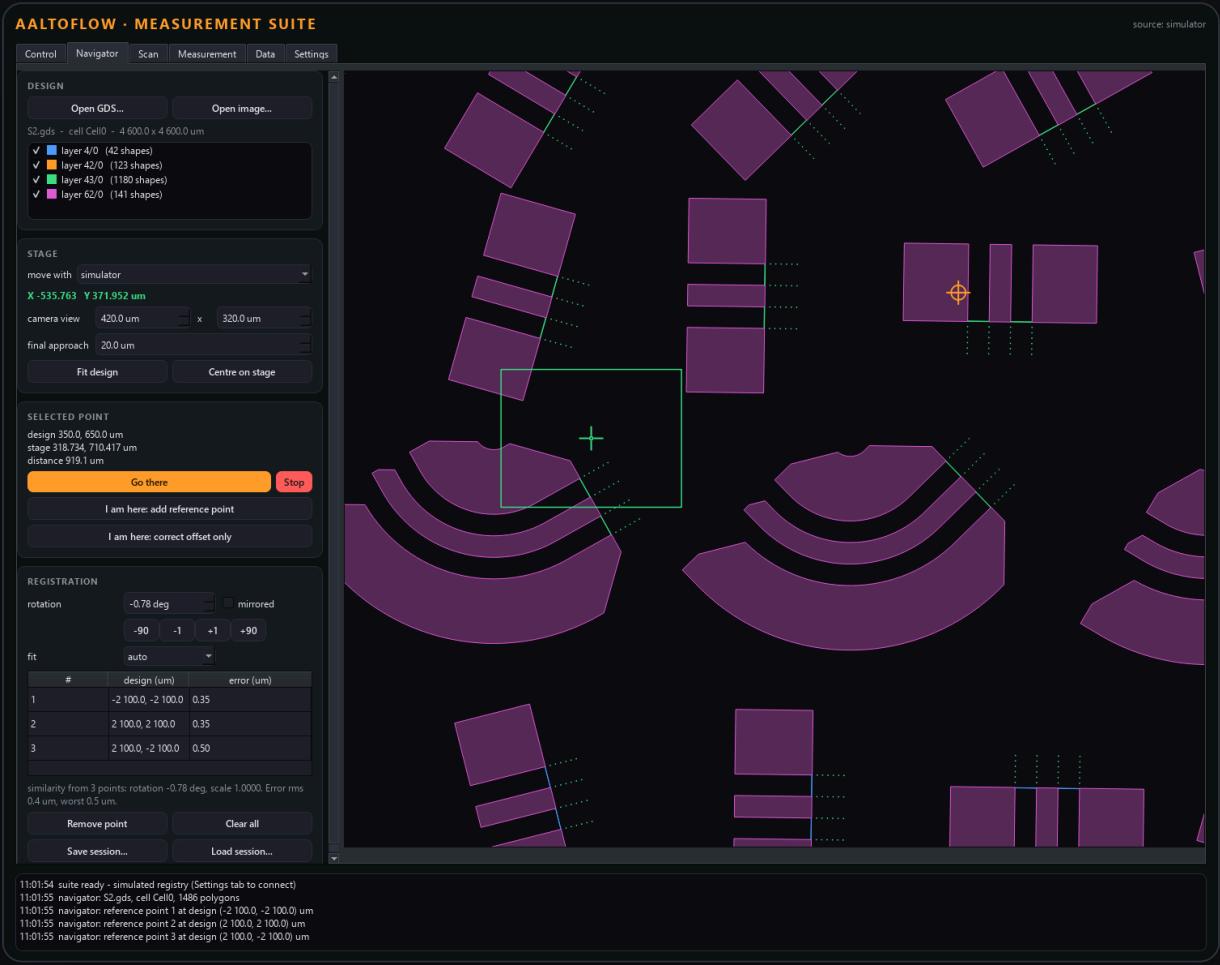
LAYOUT

A GDS file, drawn as vectors

The layout is read layer by layer and drawn as shapes, not as a picture, so it stays sharp from the whole chip down to single structures. Layers can be switched off. The dashed lines are the stage travel, the green crosshair and box are the stage position and the camera's field of view, the amber mark is the selected point, and the diamonds are the reference points.



A 4.6 mm chip layout registered with two reference points; the selected point is 2.7 mm from the stage position



Three reference points, errors 0.35–0.50 μm ; zoomed in on the layout around the camera's field of view

REGISTER

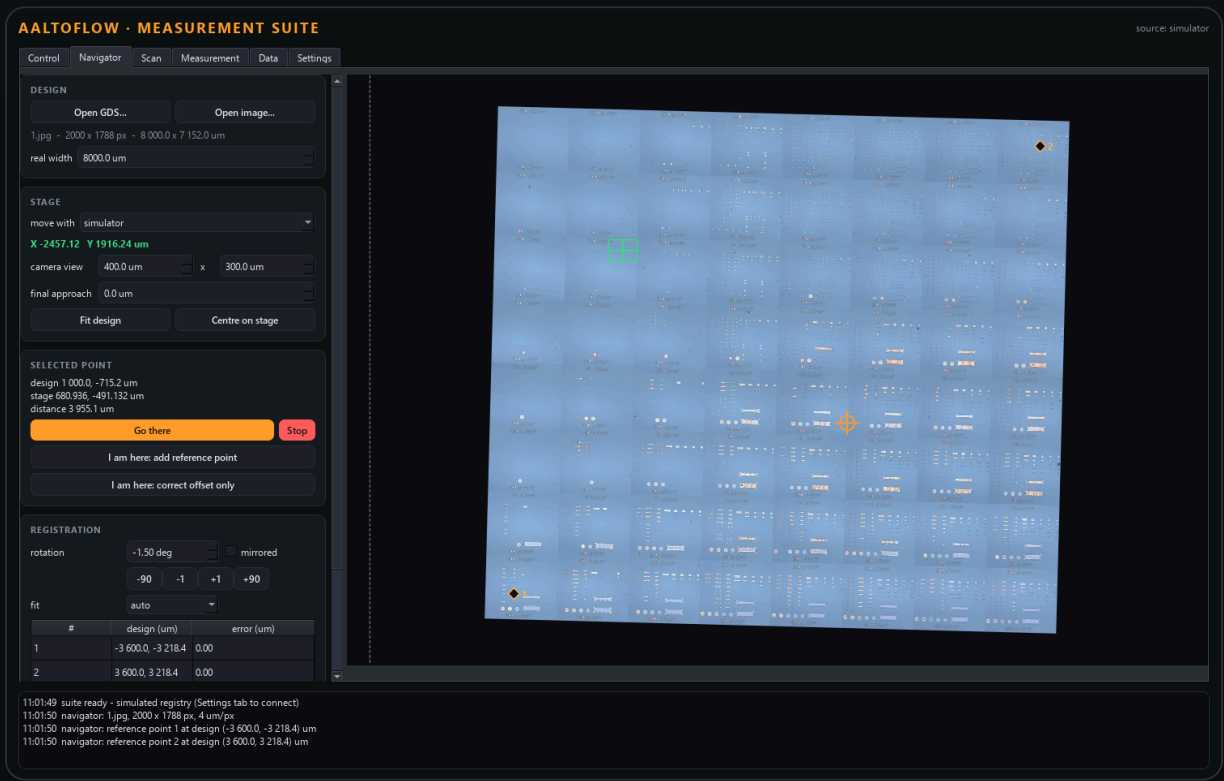
Reference points: “I am here”

Put a feature under the laser, click the same feature on the layout and press “I am here”. One point sets the position and uses the rotation set by eye. A second point fits the rotation and the stage's scale; the layout's scale is exact, so a fitted scale different from 1 is the stage's error. From three points each point's error is listed and a mirrored sample is detected; from four, X and Y may scale differently.

IMAGE

A photo or a stitched micrograph

Without a layout, any image works: its real width sets the scale, and the same reference points place it on the stage. A stitched overview taken on another microscope can be used to find the same structures here.



A stitched micrograph of a dose test, registered with two points

Click and go

A click shows the layout and stage coordinates, the distance and whether the point is inside the travel. “Go there” moves X, then Y, each waiting until the stage reports it has arrived.

Drift correction

An open-loop stage loses its position over long moves. “Correct offset only” shifts the mapping to where the stage really is and keeps the rotation and scale.

Final approach

Every move can end with a short step in +X and +Y, so an inertia stage arrives from the same side each time.

Any stage

The stage is found by its X and Y position controls in the module's description, in its own unit; no stage-specific code.

Sessions

The layout path, reference points and settings are saved next to the layout and loaded again the next day.

Safe with scans

No move while a scan is running; targets outside the travel are refused.

PART OF AALTOFLOW

AaltoFlow

In AaltoFlow each instrument runs as its own service with its own window. A launcher starts them, and a scan engine drives any combination of them through multi-dimensional measurements.

github.com/FlashLukas/AaltoFlow

[scan-core](#) on GitHub

LICENCE AND CITATION

Citation

Released under the MIT licence. If you publish data measured with it, please cite:

L. Flajšman, *AaltoFlow: lab automation built from independent instrument modules*, NanoSpin group, Aalto University, doi:10.5281/zenodo.22959230

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